CS F214
Logic in CS
BITS Pilani, Hyderabad Campus
Assignment -1
Due Date : 17th October 2025 (by Midnight)
Total Marks: 30 (weightage : 10%)

Objective:

The objective is to implement storing a propositional logic formula using a parse tree and run various algorithms you have learnt in this course on this representation of a propositional logic formula.

As first step you have to implement the construction of a parse tree for a propositional logic formula. You can assume that the formula is well formed and fully parenthesized. The input to your program should be a propositional logical formula in infix notation. The connective symbols you will use for this assignment is as follows.

1. ~ for negation
2. + for OR
3. * for AND
4. > for implication.

Your code should have **separate functions** to perform the given tasks.

Task 1:
Write a function to **convert the infix** propositional logic expression **into a prefix** propositional logic expression. **[2]**

Task 2:
Write a function to **convert the prefix expression into a rooted binary parse tree**. **[3]**

Task 3:
Write a function to traverse the parse tree to **output the infix expression** back by in-order traversal of the parse tree. **[2]**

Task 4:
Write a function to **compute the height of a parse tree.** **[3]**

Task 5:
Write a function to **evaluate the truth value of a propositional logic formula**, given the truth values of each propositional atom by traversing the tree in a bottom up fashion. **[3]**

Task 6: Implement the algorithm learnt in the class to convert any propositional formula into its equivalent **CNF form**. The input to your algorithm is a parse tree of a propositional logic formula. **[7]**

Task 7: Implement the validity check algorithm on the CNF formula to check if it is valid or not. **[4]**

Task 8:
Analyze your algorithms in terms of output, efficiency, and memory usage. Make sure that your code is well indented and commented. Document your code using Doxygen. Document your results and analysis as HTML pages (pay attention to the aesthetics).

You can use the below dataset(s) for all the tasks (except task 7). **[6]**

Data Set: Refer to the SAT competition website:
https://satcompetition.github.io/

Pickup SAT 2002 benchmark. Download the compressed folder. (If you have download issues, please contact me).

You will find the files are in .cnf format. These are text files and hence can be opened using text editors.

Here is a website explaining the .cnf file format

https://jix.github.io/varisat/manual/0.2.0/formats/dimacs.html

General Instructions:

1. This assignment will be done in groups of max five students.
2. **Code must be written in C or C++ only.**
3. There will be only one submission per group on the CMS.
4. The name of the file should be **id1_LOGIC_A1.zip**, where id1 refers to the BITS ID of the sender.
5. **You can discuss with your friends but refrain from copying the code and submitting.**
6. You have to demo the code to the instructor on a scheduled date and timing after submission. **It is important to attend the demo, as absence from demo will amount to no credit for the assignment. The tentative date for the Demo will be right after the Midsem exam.**

References to look into:
http://scanftree.com/Data_Structure/infix-to-prefix

Algorithms in C++ by Robert Sedgewick